

Occam-Decoherence: Parsimony as a Physical Thermodynamic Limit

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Abstract

Occam's Razor is heavily utilized as a philosophical heuristic in cosmological formulation. We redefine it entirely as a hard mathematical and information-processing limit constraints: Cosmological Parsimony. This paper constructs a mathematical framework demonstrating that physical space natively forces wave-function collapse and state deletion when localized reality arrays exceed defined maximum computation densities. I prove mathematically that nature does not maintain infinite computational branch possibilities; instead, utilizing strict algorithmic tensor limits, it triggers immediate probability reduction to preserve geometric reality structure.

Keywords: Wave-function Collapse, Occam's Razor, Matrix Computation limits, Decoherence.

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1 Algorithmic Formalization of Information Compression

Within bounded system models, local regions encode maximum entropy processing rates [Shannon \[1948\]](#). As chaotic probability chains (such as atomic entanglements geometry) interact with macro-variables, information complexity geometrically outpaces the operational grid bounds [Lamport \[1978\]](#).

Theorem 1.1 (Matrix Pruning Action). *For any physical state $\mathcal{S}$, when the Shannon description scale length $L(\mathcal{S})$ breaches the bounds K_{max} , algorithmic resolution automatically reduces the equation down to the state requiring the absolute minimum energy cost requirement to simulate.*

$$\min_E(\mathcal{S}_{reduction}) = C_0 \quad (1)$$

Occam's heuristic operates therefore not out of aesthetic choice, but out of absolute mathematically enforced universal bandwidth limits [Turing \[1936\]](#).

2 Conclusion

Physical reality operates using active Cosmological Parsimony models—crushing exponential branches of probability strictly to conserve $O(1)$ stability in a discrete system.

References

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